Software Requirement Specification (SRS)

SkillSync - AI-Powered Project Management System

1. Introduction

1.1 Purpose

The purpose of this system is to build an **AI-Powered Project Management Platform** with intelligent skill-based team assembly, automated task assignment, predictive analytics, and real-time collaboration features. The application will serve as an intelligent workspace where organizations can optimize project staffing, track capacity, and improve project success rates through data-driven insights.

1.2 Scope

The platform will provide:

- **Smart Team Assembly:** AI automatically suggests optimal team compositions based on required skills, availability, and past collaboration success rates.
- **Skill-Based Project Matching:** Projects are matched to team members based on skill proficiency, availability, and workload.
- **Predictive Analytics:** AI predicts project success probability, identifies potential bottlenecks, and suggests interventions.
- **Dynamic Task Assignment:** Tasks are automatically assigned based on skills, current workload, and capacity.
- **Real-Time Collaboration:** Team communication through comments, mentions, notifications, and activity feeds.
- **Capacity Management:** Visual dashboards showing team availability, workload distribution, and resource utilization.
- **Skill Gap Analysis:** Identifies missing skills for projects and recommends learning resources.
- **Project Brief Parser:** AI extracts requirements, skills, and timelines from uploaded project documents.

2. Technologies

- **Frontend:** React, React Router, Zustand (state management), Tailwind CSS, shadcn/ui (components), React Flow (visual graphs), Recharts (analytics)

- **Backend:** Node.js with Express (or Python with FastAPI), PostgreSQL, Redis (caching)
- **AI/ML:** OpenAI GPT-4 API, LangChain (AI workflows), Sentence Transformers (skill matching), scikit-learn (predictive models)
- **Real-Time:** Socket.io (notifications, live updates)
- **Authentication:** JWT-based authentication with bcrypt password hashing
- **File Storage:** AWS S3 or Cloudinary (project briefs, attachments)
- **Deployment:** Docker, Docker Compose, CI/CD pipeline

3. User Roles and Permissions

3.1 Team Member (Standard User)

- View assigned projects and tasks
- Update task status and log time
- Upload files and add comments to tasks
- View own skill profile and development recommendations
- Access available projects matched to their skills
- Track personal workload and capacity
- Participate in team discussions and mentions
- View project dashboards and analytics

3.2 Project Manager

- Create and configure new projects
- Use AI to assemble teams based on skill requirements
- Assign and reassign tasks to team members
- View team capacity and workload distribution
- Access project success probability scores
- Upload project briefs for AI parsing
- Monitor project health and receive AI alerts
- Approve or modify AI-suggested team compositions
- Access advanced analytics and reporting
- Manage project timelines and milestones

3.3 Admin (System Administrator)

- Manage all users (create, edit, deactivate accounts)
- Assign roles and permissions (Member, Project Manager, Admin)
- View organization-wide analytics and metrics
- Access admin dashboard with user statistics
- Manage skill taxonomy and categories
- Configure AI model parameters and thresholds
- View audit logs and system activities
- Manage organizational settings and preferences

- Export data and generate reports
- Add or remove other administrators

4. Functional Requirements

4.1 Authentication Module

- Users must register with email and password
- Password hashing using bcrypt
- JWT-based session management
- Role assignment during registration (default: Team Member)
- Password reset via email verification
- Unauthorized users cannot access protected routes
- Session timeout after 24 hours of inactivity

4.2 User Profile Module

User profiles include:

- Personal Information: Name, email, role, bio, profile picture
- Skills: Array of skills with proficiency levels (Beginner/Intermediate/Advanced/Expert)
- Availability: Working hours per week, current capacity percentage
- Project History: List of completed and ongoing projects
- Performance Metrics: On-time delivery rate, collaboration score
- Learning Recommendations: AI-suggested courses for skill gaps
- Activity Feed: Recent activities and contributions

4.3 Project Management Module

Projects contain:

- Project Name, Description, Status (Planning/Active/Completed/On Hold)
- Required Skills: Array of skills with required proficiency levels
- Timeline: Start date, end date, milestones
- Team Members: Assigned users with their roles in the project
- Tasks: List of tasks with assignments and dependencies
- AI Success Score: Predicted probability of on-time completion (0-100%)
- Budget and Resources (optional)
- Attachments: Project briefs, documents, images

Project Views:

- Kanban Board (drag-and-drop task cards)
- List View (table format)
- Calendar View (tasks on timeline)

- Gantt Chart (optional, for timeline visualization)

4.4 AI-Powered Features

4.4.1 Smart Team Assembly

- Input: Project requirements (required skills, timeline, complexity)
- AI analyzes available team members and their skills
- Calculates skill coverage percentage for each potential team
- Considers past collaboration success rates between members
- Balances workload across the organization
- Outputs: Top 3 recommended team compositions with justifications
- Project Manager can accept, modify, or reject suggestions

4.4.2 Skill-Based Project Matching

- AI calculates match score (0-100%) between user skills and project requirements
- Uses cosine similarity on skill vectors for semantic matching
- Considers skill proficiency levels and recency
- Dashboard shows "Available Projects" ranked by match score
- Users can express interest in projects
- Notifications sent when high-match projects become available

4.4.3 Dynamic Task Auto-Assignment

- When new tasks are created, AI suggests optimal assignees
- Factors: Required skills, current workload, availability, past performance
- Can be configured as automatic or manual approval
- Re-balancing triggered when team member becomes overloaded
- Task redistribution suggestions when deadlines are at risk

4.4.4 Predictive Project Analytics

- Success Probability Score: Predicts on-time completion likelihood
- Based on: Team composition, historical data, task complexity, timeline
- Risk Detection: Identifies potential bottlenecks before they occur
- Alerts sent when success probability drops below threshold (60%)
- Resource Optimization: Suggests team adjustments or timeline changes
- Learning: Model improves accuracy over time with more project data

4.4.5 Project Brief Parser

- Upload PDF/DOCX project briefs
- AI (GPT-4) extracts: Required skills, timeline, budget, deliverables
- Auto-generates project structure with milestones
- Suggests team size based on scope
- Pre-fills project creation form
- User reviews and confirms or edits extracted information

4.4.6 Skill Gap Detection & Learning Recommendations

- Compares team skills against project requirements
- Identifies missing or under-proficient skills
- Recommends specific courses (Udemy, Coursera, LinkedIn Learning)
- Prioritizes gaps based on project importance and urgency
- Tracks skill development over time
- Updates user skills automatically based on completed projects

4.5 Task Management Module

- Create tasks with: Title, description, assigned user, due date, priority
- Task Status: To Do → In Progress → In Review → Done
- Subtasks support for breaking down complex work
- Task Dependencies: Specify which tasks must be completed first
- Time Tracking: Start/stop timer, manual time entries
- Comments: Thread-based discussions on tasks
- @Mentions: Tag team members for attention
- File Attachments: Upload relevant documents or images
- Task History: Audit trail of all changes

4.6 Collaboration Module

- Real-Time Notifications: Task assignments, mentions, comments, due dates
- Activity Feed: Project-level and personal activity streams
- In-App Messaging: Direct messages and group discussions
- Notification Preferences: Configure email vs in-app notifications
- Mention System: @username to notify specific team members
- Notification History: View all past notifications

4.7 Dashboard & Reporting Module

Personal Dashboard:

- My Tasks (filtered by status, priority)
- Available Projects (skill-matched)
- Current Workload (hours this week)
- Skill Development Recommendations
- Recent Activity
- Upcoming Deadlines

Project Dashboard:

- Project Progress (% complete)
- Tasks by Status (pie chart)
- Team Workload Distribution (bar chart)

- AI Success Score with trend
- Upcoming Milestones
- Risk Alerts
- Recent Activity

Admin Dashboard:

- Total Users, Projects, Tasks
- User Profiles with Skill Matrix
- Organizational Capacity Overview
- Project Success Rates
- Resource Utilization Metrics
- Skill Gap Analysis (organization-wide)
- User Activity Logs

4.8 Search & Filtering Module

- Global Search: Across projects, tasks, users, files
- Advanced Filters: By status, priority, assignee, due date, tags
- Skill-Based Search: Find users with specific skills
- Project Search: By name, status, team members
- Saved Filters: Create and save custom filter combinations

5. Non-Functional Requirements

Performance: API responses under 2 seconds, Dashboard loads under 3 seconds, Real-time notifications delivered within 1 second

Scalability: Support 1000+ concurrent users, Handle 10,000+ projects, Horizontal scaling with load balancers

Security: JWT authentication with token expiration, Password hashing with bcrypt (10 rounds), Role-based access control (RBAC), Input validation and sanitization, SQL injection prevention, XSS protection

Reliability: Database replication for data persistence, Automated backups every 24 hours, 99.5% uptime target, Error logging and monitoring

Usability: Responsive design (mobile, tablet, desktop), Intuitive UI with consistent design patterns, Accessibility compliance (WCAG 2.1), Loading indicators for all async operations

Maintainability: Modular codebase with clear separation of concerns, Comprehensive API documentation, Unit tests for core functions, Integration tests for API endpoints

6. Database Models (PostgreSQL Schema)

6.1 Users Table

- id (UUID, Primary Key)
- email (String, Unique, Not Null)
- password_hash (String, Not Null)
- name (String, Not Null)
- role (Enum: MEMBER, PROJECT_MANAGER, ADMIN)
- bio (Text, Optional)
- profile_picture_url (String, Optional)
- availability_hours_per_week (Integer, Default: 40)
- current_capacity_percentage (Integer, 0-100)
- created_at (Timestamp)
- updated_at (Timestamp)

6.2 Skills Table

- id (UUID, Primary Key)
- name (String, Unique, Not Null) - e.g., "React", "Python", "Project Management"
- category (String) - e.g., "Frontend", "Backend", "Soft Skills"
- created_at (Timestamp)

6.3 UserSkills Table (Junction Table)

- id (UUID, Primary Key)
- user_id (UUID, Foreign Key → Users)
- skill_id (UUID, Foreign Key → Skills)
- proficiency_level (Enum: BEGINNER, INTERMEDIATE, ADVANCED, EXPERT)
- verified (Boolean, Default: False) - admin can verify skills
- last_used (Timestamp)
- created_at (Timestamp)

6.4 Projects Table

- id (UUID, Primary Key)
- name (String, Not Null)
- description (Text)
- status (Enum: PLANNING, ACTIVE, COMPLETED, ON_HOLD)
- start_date (Date)
- end_date (Date)
- budget (Decimal, Optional)
- created_by (UUID, Foreign Key → Users)
- ai_success_score (Integer, 0-100, Default: NULL)
- created_at (Timestamp)
- updated_at (Timestamp)

6.5 ProjectSkills Table (Required Skills for Projects)

- id (UUID, Primary Key)
- project_id (UUID, Foreign Key → Projects)
- skill_id (UUID, Foreign Key → Skills)
- required_proficiency (Enum: BEGINNER, INTERMEDIATE, ADVANCED, EXPERT)
- is_critical (Boolean, Default: False)
- created_at (Timestamp)

6.6 ProjectMembers Table (Team Assignments)

- id (UUID, Primary Key)
- project_id (UUID, Foreign Key → Projects)
- user_id (UUID, Foreign Key → Users)
- role_in_project (String) - e.g., "Frontend Developer", "Team Lead"
- assigned_at (Timestamp)

6.7 Tasks Table

- id (UUID, Primary Key)
- project_id (UUID, Foreign Key → Projects)
- title (String, Not Null)
- description (Text)
- assigned_to (UUID, Foreign Key → Users, Optional)
- status (Enum: TODO, IN_PROGRESS, IN_REVIEW, DONE)
- priority (Enum: LOW, MEDIUM, HIGH, URGENT)
- due_date (Timestamp, Optional)
- estimated_hours (Decimal, Optional)
- actual_hours (Decimal, Default: 0)
- parent_task_id (UUID, Foreign Key → Tasks, Optional) - for subtasks
- created_by (UUID, Foreign Key → Users)
- created_at (Timestamp)
- updated_at (Timestamp)

6.8 TaskDependencies Table

- id (UUID, Primary Key)
- task_id (UUID, Foreign Key → Tasks) - the dependent task
- depends_on_task_id (UUID, Foreign Key → Tasks) - the task it depends on
- created_at (Timestamp)

6.9 Comments Table

- id (UUID, Primary Key)
- task_id (UUID, Foreign Key → Tasks)
- user_id (UUID, Foreign Key → Users)
- content (Text, Not Null)

- created_at (Timestamp)
- updated_at (Timestamp)

6.10 Notifications Table

- id (UUID, Primary Key)
- user_id (UUID, Foreign Key → Users)
- type (Enum: TASK_ASSIGNED, MENTION, COMMENT, DEADLINE, ALERT)
- title (String)
- message (Text)
- link (String) - URL to relevant resource
- is_read (Boolean, Default: False)
- created_at (Timestamp)

6.11 SkillMatchScores Table (Cached Match Results)

- id (UUID, Primary Key)
- user_id (UUID, Foreign Key → Users)
- project_id (UUID, Foreign Key → Projects)
- match_score (Integer, 0-100)
- skill_coverage_percentage (Integer, 0-100)
- availability_fit (Integer, 0-100)
- workload_fit (Integer, 0-100)
- calculated_at (Timestamp)

6.12 LearningRecommendations Table

- id (UUID, Primary Key)
- user_id (UUID, Foreign Key → Users)
- skill_id (UUID, Foreign Key → Skills)
- current_level (Enum: NONE, BEGINNER, INTERMEDIATE, ADVANCED)
- target_level (Enum: BEGINNER, INTERMEDIATE, ADVANCED, EXPERT)
- reason (Text) - why this skill is recommended
- course_url (String, Optional)
- course_name (String, Optional)
- priority (Enum: LOW, MEDIUM, HIGH)
- created_at (Timestamp)

7. Key API Endpoints

7.1 Authentication APIs

- POST /api/auth/register - User registration
- POST /api/auth/login - User login (returns JWT)
- POST /api/auth/logout - Logout
- POST /api/auth/forgot-password - Password reset request

- POST /api/auth/reset-password - Reset password with token

7.2 Users APIs

- GET /api/users/me - Get current user profile
- PUT /api/users/me - Update current user profile
- GET /api/users/:id - Get user by ID
- GET /api/users - List all users (admin only)
- PUT /api/users/:id/role - Update user role (admin only)

7.3 Skills APIs

- GET /api/skills - List all skills
- POST /api/skills - Create new skill (admin)
- POST /api/users/me/skills - Add skill to current user
- PUT /api/users/me/skills/:id - Update skill proficiency
- DELETE /api/users/me/skills/:id - Remove skill

7.4 Projects APIs

- GET /api/projects - List all projects
- GET /api/projects/:id - Get project details
- POST /api/projects - Create new project
- PUT /api/projects/:id - Update project
- DELETE /api/projects/:id - Delete project
- GET /api/projects/available - Get skill-matched available projects
- POST /api/projects/:id/team - Add team member
- DELETE /api/projects/:id/team/:userId - Remove team member

7.5 AI Features APIs

- POST /api/ai/team-assembly - Get AI team suggestions
- POST /api/ai/parse-brief - Parse uploaded project brief
- GET /api/ai/project-match/:projectId - Get user match scores
- POST /api/ai/predict-success/:projectId - Calculate success probability
- GET /api/ai/skill-gaps/:projectId - Identify skill gaps
- GET /api/ai/learning-recommendations - Get personalized learning paths

7.6 Tasks APIs

- GET /api/tasks - List tasks (with filters)
- GET /api/tasks/:id - Get task details
- POST /api/tasks - Create new task
- PUT /api/tasks/:id - Update task
- DELETE /api/tasks/:id - Delete task
- POST /api/tasks/:id/time - Log time on task
- POST /api/tasks/:id/comments - Add comment to task

7.7 Notifications APIs

- GET /api/notifications - Get user notifications
- PUT /api/notifications/:id/read - Mark as read
- PUT /api/notifications/read-all - Mark all as read
- DELETE /api/notifications/:id - Delete notification

7.8 Dashboard APIs

- GET /api/dashboard/personal - Personal dashboard data
- GET /api/dashboard/project/:id - Project dashboard data
- GET /api/dashboard/admin - Admin dashboard (admin only)

8. System Flow

8.1 User Onboarding

- 1. User registers with email and password
- 2. System creates user account with default role (Team Member)
- 3. User completes profile (adds skills, availability)
- 4. System calculates initial capacity and match scores
- 5. User redirected to personal dashboard

8.2 Project Creation & Team Assembly

- 1. Project Manager creates new project
- 2. PM enters project details and required skills
- 3. PM optionally uploads project brief for AI parsing
- 4. AI analyzes requirements and suggests team compositions
- 5. PM reviews suggestions and selects team
- 6. System sends notifications to assigned team members
- 7. AI calculates initial success probability
- 8. Project appears on team members' dashboards

8.3 Task Assignment & Execution

- 1. PM or system creates tasks within project
- 2. AI suggests optimal assignees based on skills and workload
- 3. Tasks assigned (manual or automatic)
- 4. Team members receive notifications
- 5. Members update task status as work progresses
- 6. Comments and mentions used for collaboration
- 7. Time tracked automatically or manually
- 8. Task completion updates project progress

8.4 AI Monitoring & Alerts

- 1. AI continuously monitors project metrics

- 2. Detects risks: low success score, overloaded members, missed deadlines
- 3. Sends alerts to Project Manager
- 4. Suggests interventions: rebalancing, timeline adjustment
- 5. PM takes action or dismisses alert
- 6. System logs decision for learning

8.5 Skill Development Cycle

- 1. AI identifies skill gaps for active projects
- 2. Generates learning recommendations for users
- 3. User views recommendations in dashboard
- 4. User completes courses (external)
- 5. User or admin updates skill proficiency
- 6. System recalculates match scores
- 7. User becomes eligible for new projects

9. AI Algorithms & Models

9.1 Skill Matching Algorithm

Uses cosine similarity on skill vectors:

- Convert user skills to vector: [skill1_proficiency, skill2_proficiency, ...]
- Convert project requirements to vector: [required_proficiency1, ...]
- Calculate cosine similarity: (A · B) / (||A|| × ||B||)
- Normalize to 0-100 scale
- Weight by skill criticality and recency

9.2 Team Assembly Optimization

- Input: Required skills, project complexity, timeline
- Generate candidate teams (combinations of available users)
- Score each team on: Skill coverage, workload balance, past collaboration
- Use greedy algorithm to find top 3 teams
- Return with justifications and alternative suggestions

9.3 Success Prediction Model

- Features: Team size, skill coverage %, avg experience, timeline, complexity
- Historical data: Past projects with success/failure outcomes
- Model: Random Forest Classifier (scikit-learn)
- Outputs: Success probability (0-100%), risk factors
- Retrains monthly with new project data

9.4 Project Brief Parser

- Uses OpenAI GPT-4 API with structured output

- Prompt: "Extract required skills, timeline, budget, deliverables from this brief"
- Response format: JSON with predefined schema
- Validates extracted data against database constraints
- Confidence scoring for each extracted field

10. Security Considerations

- Password Requirements: Minimum 8 characters, at least 1 uppercase, 1 lowercase, 1 number
- JWT Tokens: 24-hour expiration, stored in httpOnly cookies
- Rate Limiting: 100 requests per minute per IP
- Input Validation: Server-side validation on all inputs
- SQL Injection Prevention: Parameterized queries only
- XSS Prevention: Content Security Policy headers, output encoding
- CORS: Whitelist approved frontend domains only
- File Uploads: Size limits (10MB), type restrictions, virus scanning
- Audit Logging: Log all sensitive operations (role changes, deletions)
- HTTPS Only: Enforce encrypted connections in production

11. Testing Strategy

Unit Tests: Test individual functions: skill matching, score calculations, validators - Target: 80% code coverage

Integration Tests: Test API endpoints: authentication, CRUD operations, AI features - Use Jest + Supertest

E2E Tests: Test user flows: registration, project creation, task assignment - Use Cypress or Playwright

Performance Tests: Load testing with 1000+ concurrent users - Use Apache JMeter or k6

Security Tests: Penetration testing, vulnerability scanning - Use OWASP ZAP

AI Model Tests: Validate prediction accuracy, test edge cases - Compare with baseline models

12. Deployment Architecture

- Frontend: React SPA deployed on Vercel or Netlify
- Backend: Node.js/Express containerized with Docker
- Database: PostgreSQL with read replicas for scaling
- Cache: Redis for session storage and API response caching
- File Storage: AWS S3 for project briefs and attachments

- CDN: CloudFront or Cloudflare for static assets
- Load Balancer: Nginx or AWS ELB for traffic distribution
- CI/CD: GitHub Actions for automated testing and deployment
- Monitoring: Sentry (error tracking), DataDog (performance)
- Logging: ELK stack (Elasticsearch, Logstash, Kibana)

13. Future Enhancements (Post-MVP)

- Mobile App: React Native or Flutter for iOS/Android
- Integrations: Slack, Microsoft Teams, Jira, GitHub
- Advanced Analytics: Custom reports, data exports, charts
- Resource Planning: Budget tracking, expense management
- Time Off Management: PTO calendar, availability updates
- Gamification: Badges, leaderboards for task completion
- AI Chatbot: Natural language project queries
- Video Conferencing: Integrated Zoom/Google Meet
- Document Collaboration: Real-time editing like Google Docs
- Multi-language Support: Internationalization (i18n)

14. Conclusion

SkillSync represents a modern approach to project management that leverages AI to optimize team composition, predict project outcomes, and intelligently allocate resources. By automating tedious staffing decisions and providing data-driven insights, the platform empowers organizations to deliver projects more efficiently and with higher success rates. This SRS provides a comprehensive blueprint for building an MVP that can be iteratively enhanced based on user feedback and evolving organizational needs.